

Tobacco Enterprise Analytics Dashboard

Production & Sales Efficiency

Sales Performance, Production Efficiency & Target Achievement Analysis using Power BI

Power BI • DAX • Power Query • Data Modeling • Star Schema • KPI Development • Sales Analytics • Manufacturing Analytics • Business Intelligence • Data Visualization

Project Overview

This project analyzes sales performance, production efficiency, and target achievement for a tobacco manufacturing company using Power BI. The objective was to create a centralized analytics solution that enables management to monitor business performance, identify operational inefficiencies, and support data-driven decision-making.

The dashboard consolidates sales, production, distributor, and target data into a single reporting platform, providing real-time visibility into revenue growth, quality performance, production losses, and business targets.

Business Problem

The organization lacked a centralized reporting solution to monitor sales growth, production quality, waste loss, and distributor performance. Decision-makers had limited visibility into operational inefficiencies and overall business performance, making it difficult to identify growth opportunities and improve manufacturing efficiency.

This project was developed to provide a single source of truth for monitoring both commercial and operational performance.

Executive Summary

Analysis of the Tobacco Enterprise Analytics dashboard revealed the following key outcomes:

15M PKR Total Sales	6M PKR Sales YTD	1.39 Target Achievement Ratio	20.26K Units Production Waste Loss
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Sales grew from 4.4M PKR in 2024 to 5.5M PKR in 2025 and 5.9M PKR in 2026 — a 34.1% increase over three years. Monthly production rejection rates fluctuated between 3.3% and 5.0%, and total production waste reached 20.26K units, highlighting opportunities for operational improvement and cost reduction.

Key Business Metrics

KPI	Value
Total Sales	15M PKR
Sales YTD	6M PKR
Target Achievement Ratio	1.39
Production Waste Loss	20.26K Units
Sales Growth (2024–2026)	34.1%
Monthly Target Achievement	1,671% of Monthly Target (Historical Reference)
Annual Sales Target	31M PKR

Sales Performance by Distributor

Distributor	Total Sales
Punjab Corp	3.9M PKR — Top Performer
Arifwala Sales	3.3M PKR
Sahiwal Traders	2.3M PKR
Multan Mart	2.1M PKR
Cherab Dist	1.5M PKR — Lowest Performer

Punjab Corp and Arifwala Sales together contributed approximately 55% of total distributor revenue, indicating a revenue concentration risk. A significant performance gap exists between Punjab Corp (3.9M PKR) and Cherab Dist (1.5M PKR), signaling opportunities for distributor development.

Brand Portfolio

The dashboard tracks the following four brands, each filterable via interactive slicers:

- Capstan
- Diplomat
- Gold Leaf
- Morven

Revenue Growth Trend

Year	Total Sales
2024	4.4M PKR
2025	5.5M PKR
2026	5.9M PKR (YTD: 6M PKR)

The consistent upward trend reflects 34.1% growth over three years, with the strongest acceleration observed between 2025 and 2026.

Production Quality & Waste Analysis

Monthly production rejection rates fluctuated between 3.3% and 5.0% across the reporting period, indicating variability in manufacturing quality. Key observations:

- Rejection rate peaked at approximately 5.0% in months 5 and 12
- Rejection rate dipped to approximately 3.3% in months 3 and 10
- Total production waste: 20.26K units

Sales vs. Annual Target Achievement

The gauge chart in the dashboard shows actual sales of 15M PKR against an annual target of 31M PKR, with a monthly reference floor of 917K PKR. The business has achieved 1,671% of its monthly target (historical reference basis), and a Target Achievement Ratio of 1.39 for the period analyzed.

Data Model

A Star Schema data model was implemented to improve performance, scalability, and reporting accuracy. The model resolves granularity differences between monthly targets and daily sales transactions, ensuring accurate KPI calculations and time-based analysis.

Fact Tables

- Fact_Sales
- Fact_Production
- Fact_Targets

Dimension Tables

- Dim_Date
- Dim_Brand

Power BI Features Implemented

DAX Measures

- Total Sales
- Sales YTD
- Target Achievement Ratio
- Production Waste Loss
- Time Intelligence Calculations

Visualizations

- KPI Cards (Total Sales, Sales YTD, Target Achievement Ratio, Production Waste Loss)
- Area Chart: Total Sales Trend by Year (2024–2026)
- Line Chart: Monthly Production Rejection Rate Trend
- Horizontal Bar Chart: Sales Performance by Distributor
- Gauge Chart: Sales vs. Annual Target Achievement

User Experience

- Interactive Slicers: Year and Brand filters
- Cross-Filtering across all visuals
- Report Page Tooltips
- Drill-Down Navigation

Sample DAX Measures

Total Sales

```
Total Sales = SUM(Fact_Sales[SalesAmount])
```

Sales YTD

```
Sales YTD = TOTALYTD( [Total Sales], Dim_Date[Date] )
```

Target Achievement Ratio

```
Target Achievement Ratio = DIVIDE( [Total Sales], [Target Sales] )
```

Analytical Insights

Insight Area	Finding
Revenue Growth	Sales grew from 4.4M PKR (2024) to 5.9M PKR (2026), a 34.1% increase over three years.
Distributor Performance	Punjab Corp led at 3.9M PKR; Cherab Dist was lowest at 1.5M PKR.
Revenue Concentration	Top two distributors (Punjab Corp + Arifwala Sales) contributed ~55% of total revenue — concentration risk.
Production Waste	20.26K units of waste identified; opportunities for cost reduction and process optimization exist.
Product Quality	Monthly rejection rates fluctuated 3.3%–5.0%, indicating manufacturing variability.
Target Performance	Target Achievement Ratio of 1.39 demonstrates strong sales execution.
Distributor Gap	Significant gap between highest (Punjab Corp: 3.9M) and lowest (Cherab Dist: 1.5M) performers.

Business Recommendations

Quality Improvement

Conduct audits during high-rejection months and implement machine calibration procedures to reduce monthly rejection rates below 3.3%.

Distributor Development

Introduce targeted incentive programs and performance reviews for lower-performing distributors, particularly Cherab Dist and Multan Mart.

Waste Reduction

Establish production waste thresholds and implement continuous monitoring to reduce the current 20.26K units of operational waste.

Inventory Optimization

Align procurement and inventory planning with observed sales demand patterns (strong 2025–2026 growth trajectory).

Predictive Maintenance

Implement preventive maintenance schedules to minimize downtime and production disruptions that contribute to quality variability.

Brand Growth Strategy

Develop promotional campaigns for lower-performing brands to improve market penetration across all four product lines.

Workforce Training

Provide targeted training programs focused on reducing rejection rates and improving operational consistency during high-rejection months.

Strategic Planning

Use dashboard insights to align the 31M PKR annual target with realistic monthly milestones and support long-term business planning.

Projected Business Impact

Implementing the proposed recommendations could potentially:

- Reduce production waste by up to 15%
- Increase distributor revenue by approximately 10%
- Improve manufacturing efficiency and quality consistency
- Strengthen sales forecasting accuracy
- Increase overall profitability by an estimated 8%

Repository Structure

```
Dashboard.pbix  
Dataset/  
Screenshots/  
Documentation/  
README.md
```

Conclusion

This project demonstrates how Power BI can transform raw sales and production data into actionable business intelligence. By consolidating 15M PKR in sales across five distributors and four brands into a unified reporting platform, the dashboard gives management real-time visibility into revenue growth (34.1% over three years), production quality (3.3%–5.0% monthly rejection rates), waste performance (20.26K units), and target achievement (ratio of 1.39).

The Star Schema data model and DAX measures ensure accurate, scalable KPI tracking, while interactive slicers and cross-filtering empower management to drill into specific periods, brands, or distributors. The insights and recommendations delivered through this dashboard position the organization to reduce waste, develop distributor performance, and improve overall profitability.